

Signals, Systems and Transforms

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Chapter 3

Discrete-Time Signals and Systems

Chapter Intended Learning Outcomes:

- i) Understanding deterministic and random discrete-time signals and ability to generate them
- ii) Ability to recognize the discrete-time system properties, namely, memorylessness, stability, causality, linearity and time-invariance
- iii) Understanding discrete-time convolution and ability to perform its computation
- iv) Understanding the relationship between difference equations and discrete-time signals and systems

Discrete-Time Signal

- Discrete-time signal can be generated using a computing software such as MATLAB/OCTAVE
- It can also be obtained from **sampling** continuous-time signals in real world

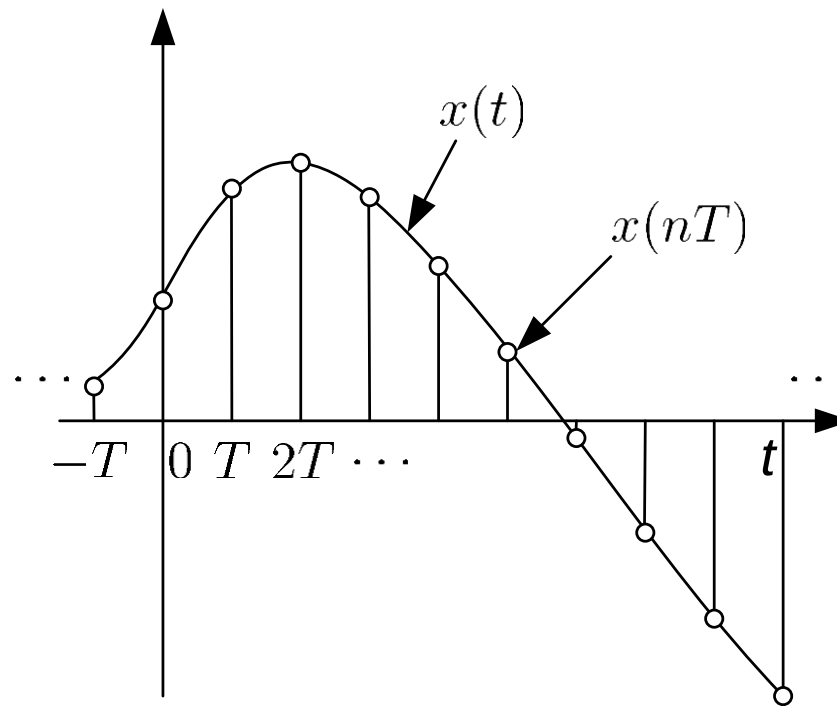


Fig.3.1 Discrete-time signal obtained from analog signal

- The discrete-time signal $x(nT)$ is equal to $x(t)$ only at the sampling interval of $t = nT$, $n = \dots - 1, 0, 1, 2, \dots$

$$x[n] = x(t)|_{t=nT} = x(nT), \quad n = \dots - 1, 0, 1, 2, \dots \quad (3.1)$$

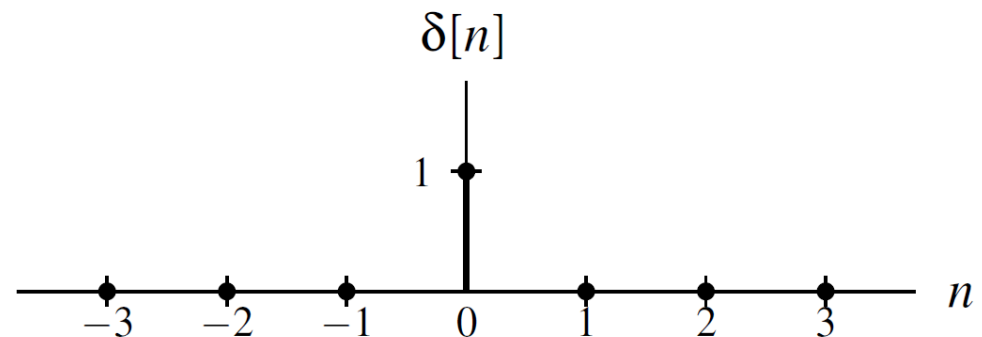
where T is called the **sampling period**

- $x[n]$ is a **sequence of numbers**, $\dots x[-1], x[0], x[1], x[2], \dots$, with n being the time index

Basic Sequences

- **Unit Sample (or Impulse)**

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} \quad (3.2)$$

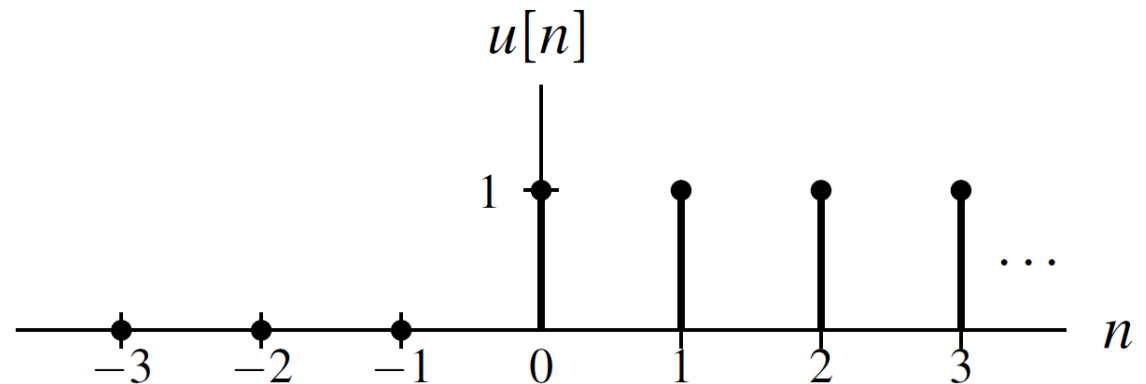


It is similar to the continuous-time unit $\delta(t)$ which is impulse defined in Chapter 2.

$\delta[n]$ is simpler than $\delta(t)$ because it is well defined for all n while $\delta(t)$ is not defined at $t = 0$

■ Unit Step

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad (3.3)$$



It is similar to the continuous-time $u(t)$ (see Chap 2)

$\delta[n]$ is an important function because it serves as the building block of any discrete-time signal $x[n]$:

$$\begin{aligned} x[n] &= \cdots + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + \cdots \\ &= \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \end{aligned} \quad (3.4)$$

For example, $u[n]$ can be expressed in terms of $\delta[n]$ as:

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k] \quad (3.5)$$

Conversely, we can use $u[n]$ to represent $\delta[n]$:

$$\delta[n] = u[n] - u[n-1] \quad (3.6)$$

Discrete-Time Signal Generation

A **deterministic** discrete-time signal $x[n]$ satisfies a generating model with **known** functional form:

$$x[n] = f(\boldsymbol{\psi}, n) \quad (3.7)$$

where $f(\cdot)$ is a function of **parameter** vector $\boldsymbol{\psi}$ and **time index** n . That is, given $f(\cdot)$ and $\boldsymbol{\psi}$, $x[n]$ can be produced

e.g., the time-shifted unit sample $\delta[n - n_0]$ and unit step function $u[n - n_0]$, where the parameter is n_0

e.g., for an exponential function α^{n-n_0} , we have $\boldsymbol{\psi} = [\alpha \quad n_0]$ where α is the decay factor and n_0 is the time shift

e.g., for a sinusoid $A \cos(\omega n + \theta)$, we have $\boldsymbol{\psi} = [A \quad \omega \quad \theta]$

Example 3.1

Use MAT/OCT to generate a discrete-time sinusoid of the form:

$$x[n] = A \cos(\omega n + \theta), \quad n = 0, 1, \dots, N - 1$$

with $A = 1$, $\omega = 0.3$, $\theta = 1$ and $N = 21$, which has a duration of 21 samples

We can generate $x[n]$ by using the following MAT/OCT code:

```
N=21;           %number of samples is 21
A=1;            %tone amplitude is 1
w=0.3;          %frequency is 0.3
p=1;            %phase is 1
for n=1:N
x(n)=A*cos(w*(n-1)+p); %time index should be >0
end
```

Note that x is a vector and its index should be at least 1.

Alternatively, we can also use:

```
N=21;           %number of samples is 21 %tone
A=1;           amplitude is 1
w=0.3;         %frequency is 0.3
p=1;           %phase is 1
n=0:N-1;       %define time index vector %first
               time index is also 1
x=A.*cos(w.*n+p);
```

To plot $x[n]$, we can either use the commands `stem(x)` and `plot(x)`

If the time index is not specified, the default start time is $n = 1$

Nevertheless, it is easy to include the time index vector in the plotting command

e.g., Using `stem` to plot $x[n]$ with the correct time index:

```
n=0:N-1;       % n is vector of time index
stem(n,x)      % plot x versus n
```

Similarly, `plot(n,x)` can be employed to show $x[n]$

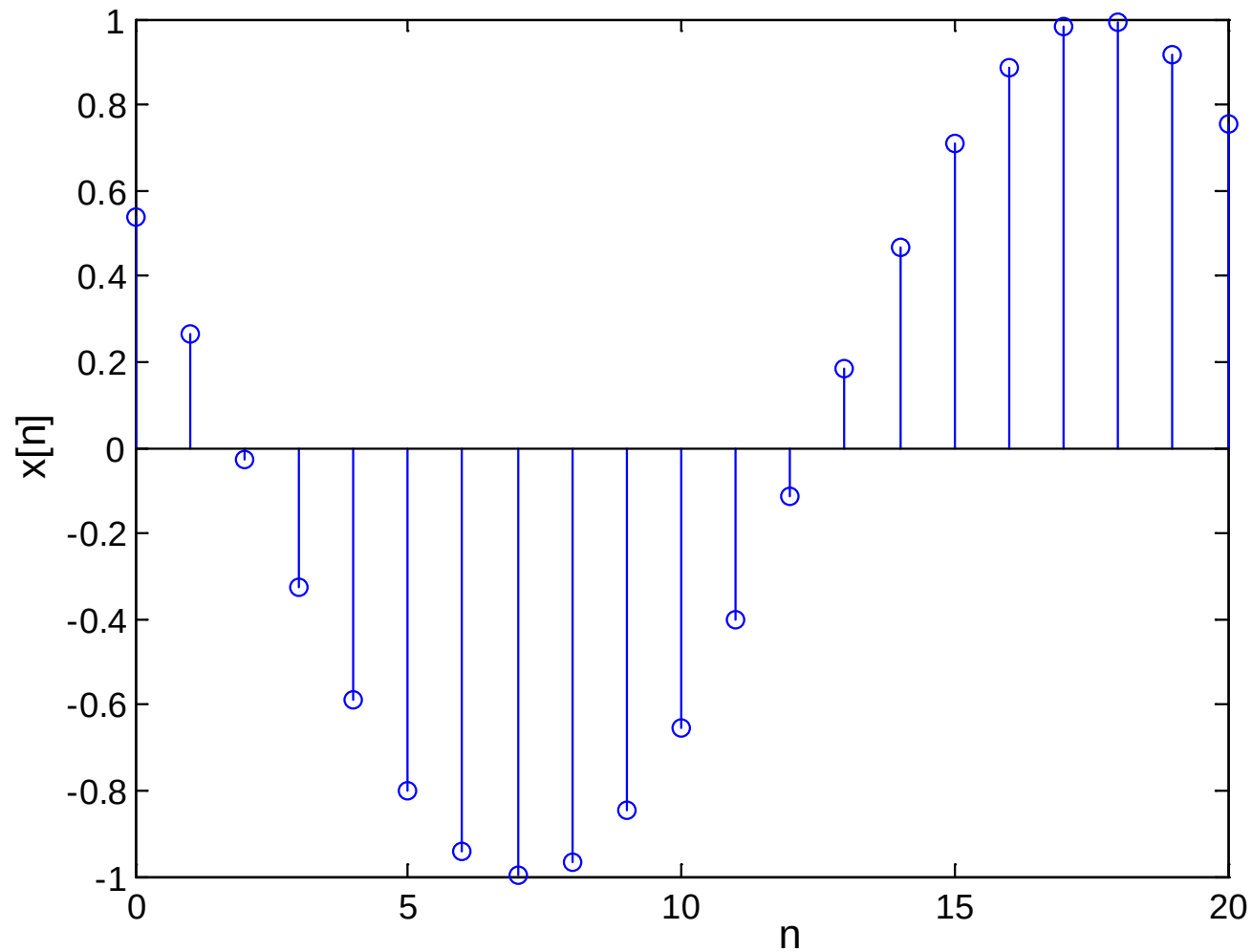


Fig.3.2 Plot of discrete-time sinusoid using stem

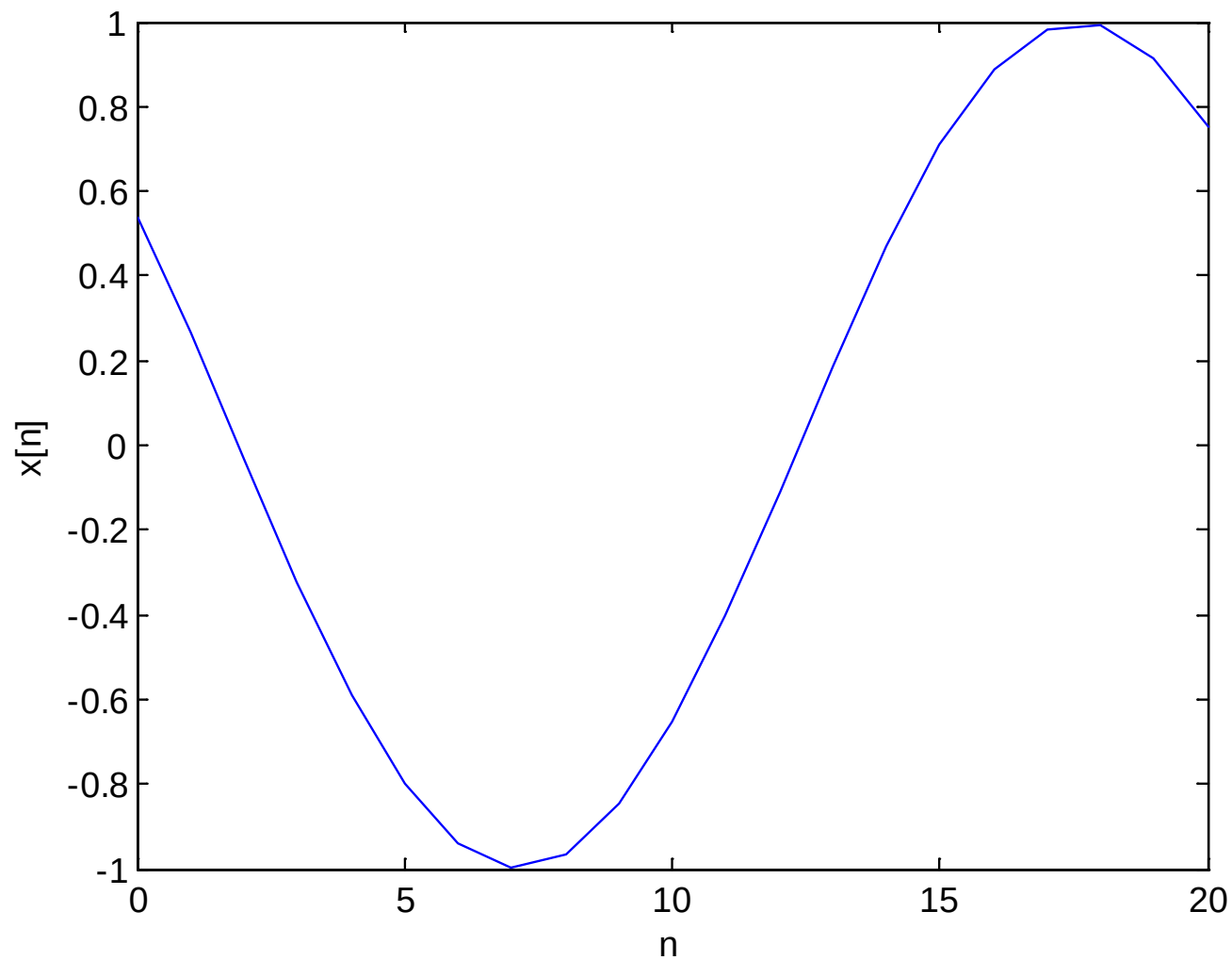


Fig.3.3 Plot of discrete-time sinusoid using plot

Discrete-Time Systems

A discrete-time system is an operator $\mathcal{T}$ which maps an input sequence $x[n]$ into an output sequence $y[n]$:

$$y[n] = \mathcal{T}\{x[n]\} \quad (3.8)$$

- **Memoryless**: $y[n]$ at time n depends **only** on $x[n]$ at time n

A memoryless system's output is determined by the current input value only, hence, every memoryless system must also be causal (a system is causal if its output does not depend on the future input values). The converse in general is not true, however: a causal system can have memory (depend on past input), or can be memoryless (depend on current input).

- **Linear**: obey principle of **superposition**, i.e., if

$$y_1[n] = \mathcal{T}\{x_1[n]\} \quad \text{and} \quad y_2[n] = \mathcal{T}\{x_2[n]\}$$

then

$$\mathcal{T}\{ax_1[n] + bx_2[n]\} = a\mathcal{T}\{x_1[n]\} + b\mathcal{T}\{x_2[n]\} = ay_1[n] + by_2[n] \quad (3.9)$$

Example 3.2

Determine whether the following system with input $x[n]$ and output $y[n]$, is linear or not:

$$y[n] = \sum_{k=-\infty}^n x[k]$$

A standard approach to determine the linearity of a system is given as follows. Let

$$y_i[n] = \mathcal{T}\{x_i[n]\}, \quad i = 1, 2, 3$$

with

$$x_3[n] = ax_1[n] + bx_2[n]$$

If $y_3[n] = ay_1[n] + by_2[n]$, then the system is linear. Otherwise, the system is nonlinear.

Assigning $x_3[n] = ax_1[n] + bx_2[n]$, we have:

$$\begin{aligned} y_3[n] &= \sum_{k=-\infty}^n x_3[k] \\ &= \sum_{k=-\infty}^n (ax_1[k] + bx_2[k]) \\ &= a \sum_{k=-\infty}^n x_1[k] + b \sum_{k=-\infty}^n x_2[k] \\ &= ay_1[n] + by_2[n] \end{aligned}$$

Note that the outputs for $x_1[n]$ and $x_2[n]$ are $y_1[n] = \sum_{k=-\infty}^n x_1[k]$ and $y_2[n] = \sum_{k=-\infty}^n x_2[k]$

As a result, the system is linear

Example 3.3

Determine whether the following system with input $x[n]$ and output $y[n]$, is linear or not:

$$y[n] = 3x^2[n] + 2x[n - 3]$$

The system outputs for $x_1[n]$ and $x_2[n]$ are

$y_1[n] = 3x_1^2[n] + 2x_1[n - 3]$ and $y_2[n] = 3x_2^2[n] + 2x_2[n - 3]$. Assigning $x_3[n] = ax_1[n] + bx_2[n]$, its system output is then:

$$\begin{aligned} y_3[n] &= 3x_3^2[n] + 2x_3[n - 3] \\ &= 3(ax_1[n] + bx_2[n])^2 + 2ax_1[n - 3] + 2bx_2[n - 3] \\ &= 3(a^2x_1^2[n] + b^2x_2^2[n] + 2abx_1[n]x_2[n]) + 2ax_1[n - 3] + 2bx_2[n - 3] \\ &\neq a(3x_1^2[n] + 2x_1[n - 3]) + b(3x_2^2[n] + 2x_2[n - 3]) \\ &= ay_1[n] + by_2[n] \end{aligned}$$

As a result, the system is nonlinear

- **Time-Invariant:** a time-shift of input causes a corresponding shift in output, i.e., if

$$y[n] = \mathcal{T}\{x[n]\}$$

then

$$y[n - n_0] = \mathcal{T}\{x[n - n_0]\} \quad (3.10)$$

Example 3.4

Determine whether the following system with input $x[n]$ and output $y[n]$, is time-invariant or not:

$$y[n] = \sum_{k=-\infty}^n x[k]$$

A standard approach to determine the time-invariance of a system is given as follows.

Let

$$y_1[n] = \mathcal{T}\{x_1[n]\}$$

with $x_1[n] = x[n - n_0]$

If $y_1[n] = y[n - n_0]$, then the system is time-invariant.
Otherwise, the system is time-variant.

From the given input-output relationship, $y[n - n_0]$ is:

$$y[n - n_0] = \sum_{k=-\infty}^{n-n_0} x[k]$$

Let $x_1[n] = x[n - n_0]$, its system output is:

$$\begin{aligned} y_1[n] &= \sum_{k=-\infty}^n x_1[k] = \sum_{k=-\infty}^n x[k - n_0] = \sum_{l=-\infty}^{n-n_0} x[l], \quad l = k - n_0 \\ &= y[n - n_0] \end{aligned}$$

As a result, the system is time-invariant.

Example 3.5

Determine whether the following system with input $x[n]$ and output $y[n]$, is time-invariant or not:

$$y[n] = 3x[3n]$$

From the given input-output relationship, $y[n - n_0]$ is of the form:

$$y[n - n_0] = 3x[3(n - n_0)] = 3x[3n - 3n_0]$$

Let $x_1[n] = x[n - n_0]$, its system output is:

$$y_1[n] = 3x_1[3n] = 3x[3n - n_0] \neq y[n - n_0]$$

As a result, the system is time-variant.

- **Causal**: output $y[n]$ at time n depends on input $x[n]$ up to time n

For **linear time-invariant** (LTI) systems, there is an alternative definition. A LTI system is causal if its impulse response $h[n]$ satisfies:

$$h[n] = 0, \quad n < 0 \quad (3.11)$$

Are they causal systems?

$$y[n] = x[n] + x[n+1]$$

$$y[n] = x[n] + x[n-2]$$

- **Stable**: a bounded input $x[n]$ ($|x[n]| < \infty$) produces a bounded output $y[n]$ ($|y[n]| < \infty$)

For LTI system, stability also corresponds to

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty \quad (3.12)$$

Are they stable systems?

$$y[n] = x[n] + x[n+1]$$

$$y[n] = 1/x[n]$$

Convolution

The input-output relationship for a LTI system is characterized by **convolution**:

$$y[n] = x[n] \otimes h[n] = \sum_{m=-\infty}^{\infty} x[m]h[n-m] \quad (3.13)$$

which is similar to (2.23)

(3.13) is simpler as it only needs **additions** and **multiplications**

$h[n]$ specifies the functionality of the system

■ **Commutative**

$$\begin{aligned} x[n] \otimes h[n] &= h[n] \otimes x[n] \\ &= \sum_{m=-\infty}^{\infty} x[m]h[n-m] = \sum_{m=-\infty}^{\infty} h[m]x[n-m] \end{aligned} \quad (3.14)$$

and

$$\begin{aligned} y[n] &= x[n] \otimes h_1[n] \otimes h_2[n] \\ &= x[n] \otimes h_2[n] \otimes h_1[n] \\ &= x[n] \otimes (h_1[n] \otimes h_2[n]) \end{aligned} \quad (3.15)$$

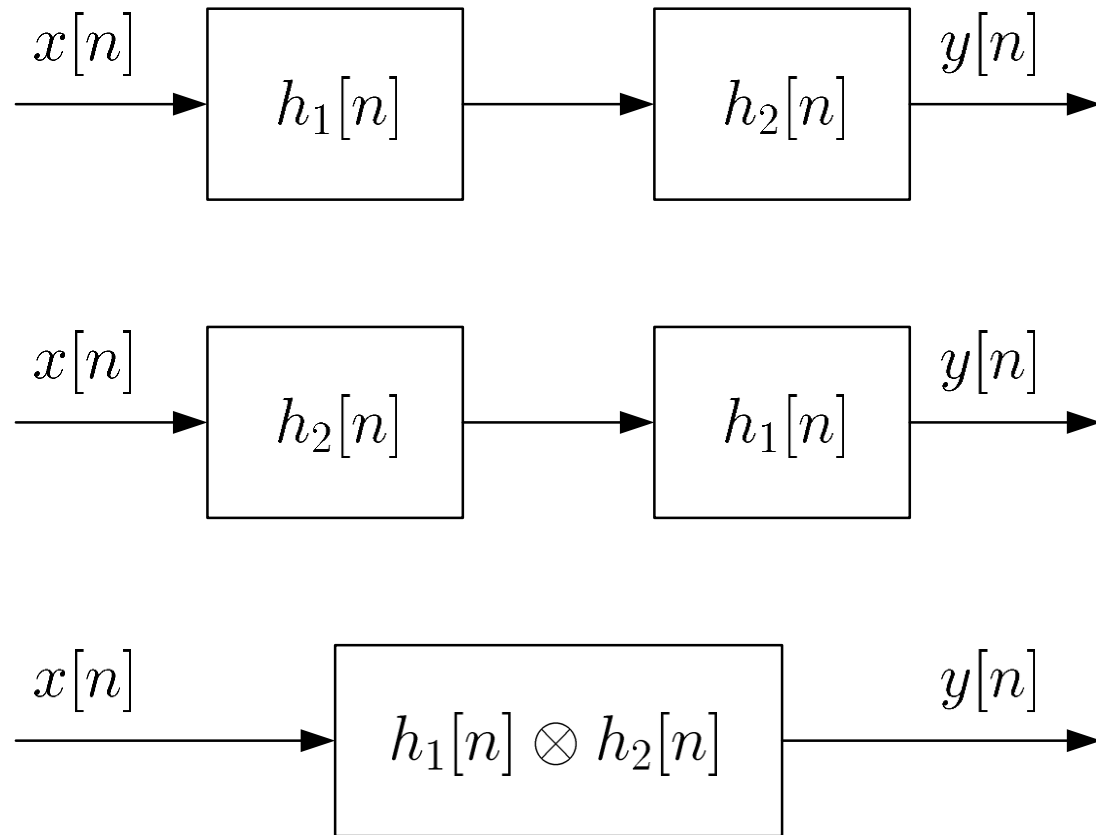


Fig.3.4 Commutative property of convolution

- Linearity

$$y[n] = x[n] \otimes (h_1[n] + h_2[n]) = x[n] \otimes h_1[n] + x[n] \otimes h_2[n] \quad (3.16)$$

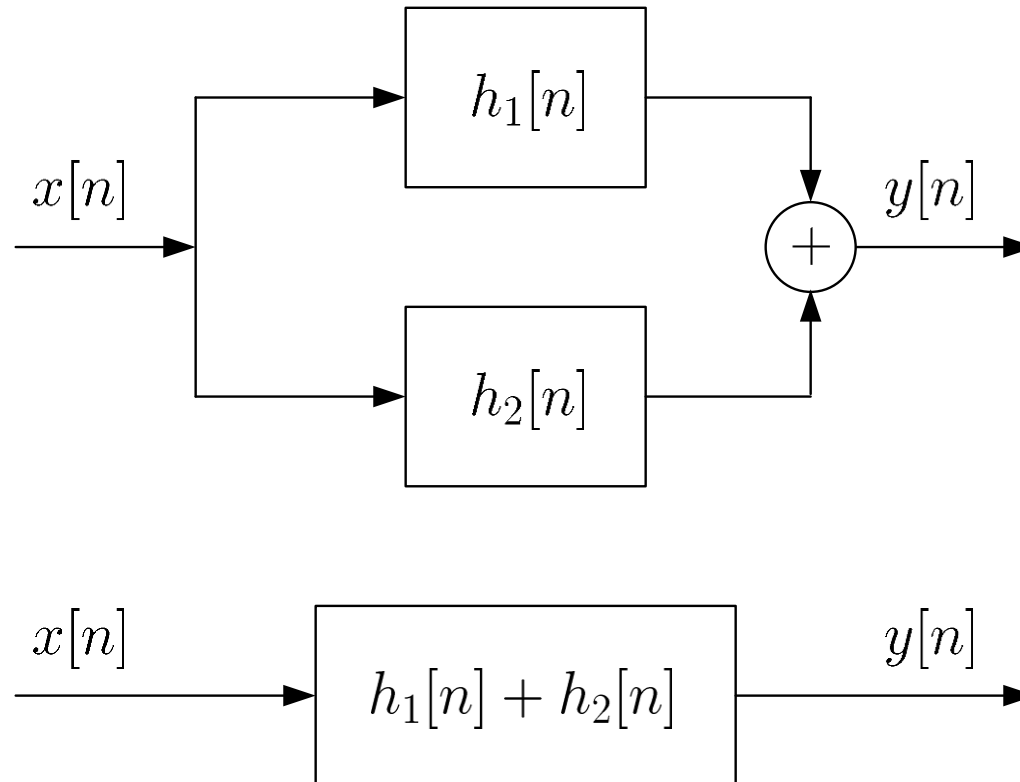


Fig.3.5 Linear property of convolution

Example 3.6

Compute the output $y[n]$ if the input is $x[n] = u[n]$ and the LTI system impulse response is $h[n] = \delta[n] + 0.5\delta[n - 1]$. Determine the stability and causality of system.

Using (3.13), we have:

$$\begin{aligned} y[n] &= x[n] \otimes h[n] = \sum_{m=-\infty}^{\infty} x[m]h[n - m] \\ &= \sum_{m=-\infty}^{\infty} u[m] (\delta[n - m] + 0.5\delta[n - 1 - m]) \\ &= \sum_{m=0}^{\infty} (\delta[n - m] + 0.5\delta[n - 1 - m]) \\ &= \sum_{m=0}^{\infty} \delta[n - m] + 0.5 \sum_{m=0}^{\infty} \delta[n - 1 - m] = u[n] + 0.5u[n - 1] \end{aligned}$$

Alternatively, we can first establish the general relationship between $y[n]$ and $x[n]$ with the specific $h[n]$ and (3.4):

$$\begin{aligned} y[n] &= x[n] \otimes h[n] = \sum_{m=-\infty}^{\infty} x[m]h[n-m] \\ &= \sum_{m=-\infty}^{\infty} x[m] (\delta[n-m] + 0.5\delta[n-1-m]) \\ &= \sum_{m=-\infty}^{\infty} x[m]\delta[n-m] + 0.5 \sum_{m=-\infty}^{\infty} x[m]\delta[n-1-m] \\ &= x[n] + 0.5x[n-1] \end{aligned}$$

Substituting $x[n] = u[n]$ yields the same $y[n]$.

Since $\sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=0}^1 |h[n]| = 1.5 < \infty$ and $h[n] = 0$ for $n < 0$ the system is stable and causal

Example 3.7

Compute the output $y[n]$ if the input is $x[n] = a^n u[n]$ and the LTI system impulse response is $h[n] = u[n] - u[n - 10]$. Determine the stability and causality of system.

Using (3.13), we have:

$$\begin{aligned} y[n] &= \sum_{m=-\infty}^{\infty} x[m] h[n - m] \\ &= \sum_{m=-\infty}^{\infty} a^m u[m] (u[n - m] - u[n - 10 - m]) \\ &= \sum_{m=0}^{\infty} a^m (u[n - m] - u[n - 10 - m]) \\ &= \sum_{m=0}^{\infty} a^m u[n - m] - \sum_{m=0}^{\infty} a^m u[n - 10 - m] \end{aligned}$$

Let $y_1[n] = \sum_{m=0}^{\infty} a^m u[n-m]$ and $y_2[n] = \sum_{m=0}^{\infty} a^m u[n-10-m]$ such that $y[n] = y_1[n] - y_2[n]$. By employing a change of variable, $y_1[n]$ is expressed as

$$\begin{aligned} y_1[n] &= \sum_{m=0}^{\infty} a^m u[n-m] = \sum_{k=n}^{-\infty} a^{n-k} u[k], \quad k = n - m \\ &= \sum_{k=-\infty}^n a^{n-k} u[k] \end{aligned}$$

Since $u[k] = 0$ for $k < 0$, $y_1[n] = 0$ for $n < 0$. For $n \geq 0$, $y_1[n]$ is:

$$y_1[n] = \sum_{k=0}^n a^{n-k} u[k] = 1 + a + \cdots + a^n = \frac{1 - a^{n+1}}{1 - a}$$

That is,

$$y_1[n] = \frac{1 - a^{n+1}}{1 - a} u[n]$$

Similarly, $y_2[n]$ is:

$$\begin{aligned} y_2[n] &= \sum_{m=0}^{\infty} a^m u[n - 10 - m] \\ &= \sum_{k=-\infty}^{n-10} a^{n-10-k} u[k], \quad k = n - 10 - m \end{aligned}$$

Since $u[k] = 0$ for $k < 0$, $y_2[n] = 0$ for $n < 10$. For $n \geq 10$, $y_2[n]$ is:

$$y_2[n] = \sum_{k=0}^{n-10} a^{n-10-k} = 1 + a + \cdots + a^{n-10} = \frac{1 - a^{n-9}}{1 - a}$$

That is,

$$y_2[n] = \frac{1 - a^{n-9}}{1 - a} u[n - 10]$$

Combining the results, we have:

$$y[n] = \frac{1 - a^{n+1}}{1 - a}u[n] - \frac{1 - a^{n-9}}{1 - a}u[n - 10]$$

or

$$y[n] = \begin{cases} 0, & n < 0 \\ \frac{1 - a^{n+1}}{1 - a}, & 0 \leq n < 10 \\ \frac{a^{n-9}(1 - a^{10})}{1 - a}, & 10 \leq n \end{cases}$$

Since $\sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=0}^9 |h[n]| = 10 < \infty$, the system is stable. Moreover, the system is causal because $h[n] = 0$ for $n < 0$.

Example 3.8

Determine $y[n] = x[n] \otimes h[n]$ where $x[n]$ and $h[n]$ are

$$x[n] = \begin{cases} n^2 + 1, & 0 \leq n \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

and

$$h[n] = \begin{cases} n + 1, & 0 \leq n \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

Here, the lengths of both $x[n]$ and $h[n]$ are finite. More precisely, $x[0] = 1$, $x[1] = 2$, $x[2] = 5$, $x[3] = 10$, $h[0] = 1$, $h[1] = 2$, $h[2] = 3$ and $h[3] = 4$ while all other $x[n]$ and $h[n]$ have zero values.

We still use (3.13) but now it reduces to a finite summation:

$$\begin{aligned} y[n] &= \sum_{m=-\infty}^{\infty} x[m]h[n-m] \\ &= x[0]h[n] + x[1]h[n-1] + x[2]h[n-2] + x[3]h[n-3] \end{aligned}$$

By considering the non-zero values of $h[n]$, we obtain:

$$y[n] = \begin{cases} 1, & n = 0 \\ 4, & n = 1 \\ 12, & n = 2 \\ 30, & n = 3 \\ 43, & n = 4 \\ 50, & n = 5 \\ 40, & n = 6 \\ 0, & \text{otherwise} \end{cases}$$

Alternatively, for finite-length discrete-time signals, we can use the MATLAB command `conv` to compute the convolution of **finite-length** sequences:

```
n=0:3;  
x=n.^2+1;  
h=n+1;  
y=conv(x,h)
```

The results are

$y = 1 \quad 4 \quad 12 \quad 30 \quad 43 \quad 50 \quad 40$

As the default starting time indices in both `h` and `x` are 1, we need to determine the appropriate time index for `y`

The correct index can be obtained by computing one value of $y[n]$ using (3.13). For simplicity, we may compute $y[0]$:

$$\begin{aligned} y[0] &= \sum_{m=-\infty}^{\infty} x[m]h[-m] \\ &= \cdots + x[-1]h[1] + x[0]h[0] + x[1]h[-1] + \cdots \\ &= x[0]h[0] \\ &= 1 \end{aligned}$$

In general, if the lengths of $x[n]$ and $h[n]$ are M and N , respectively, the length of $y[n] = x[n] \otimes h[n]$ is $(M + N - 1)$.

Linear Constant Coefficient Difference Equations

For a LTI system, its input $x[n]$ and output $y[n]$ are related via a N th-order linear constant coefficient **difference equation**:

$$\sum_{k=0}^N a_k y[n - k] = \sum_{k=0}^M b_k x[n - k] \quad (3.17)$$

which is useful to check whether a system is both linear and time-invariant or not

Example 3.9

Determine if the following input-output relationships correspond to LTI systems:

- (a) $y[n] = 0.1y[n - 1] + x[n] + x[n - 1]$
- (b) $y[n] = x[n + 1] + x[n]$
- (c) $y[n] = 1/x[n]$

We see that (a) corresponds to a LTI system with $N = M = 1$, $a_0 = 1$, $a_1 = -0.1$ and $b_0 = b_1 = 1$

For (b), we reorganize the equation as:

$$y[n] = x[n+1] + x[n] \Rightarrow y[n-1] = x[n] + x[n-1]$$

which agrees with (3.17) when $N = M = 1$, $a_0 = 0$ and $a_1 = b_0 = b_1 = 1$. Hence (b) also corresponds to a LTI system

For (c), it does not correspond to a LTI system because $x[n]$ and $y[n]$ are not linear in the equation

Note that if a system cannot be fitted into (3.17), there are three possibilities: linear and time-variant; nonlinear and time-invariant; or nonlinear and time-variant

Example 3.10

Compute the impulse response $h[n]$ for a LTI system which is characterized by the following difference equation:

$$y[n] = x[n] - x[n - 1]$$

Expanding (3.13) as

$$\begin{aligned} y[n] &= \sum_{m=-\infty}^{\infty} h[m]x[n - m] \\ &= \cdots + h[-1]x[n + 1] + h[0]x[n] + h[1]x[n - 1] + \cdots \end{aligned}$$

we can easily deduce that only $h[0]$ and $h[1]$ are nonzero. That is, the impulse response is:

$$h[n] = \delta[n] - \delta[n - 1]$$

The difference equation is also useful to generate the system output and input.

Assuming that $a_0 \neq 0$, $y[n]$ is computed as:

$$y[n] = \frac{1}{a_0} \left(- \sum_{k=1}^N a_k y[n-k] + \sum_{k=0}^M b_k x[n-k] \right) \quad (3.18)$$

Assuming that $b_0 \neq 0$, $x[n]$ can be obtained from:

$$x[n] = \frac{1}{b_0} \left(\sum_{k=0}^N a_k y[n-k] - \sum_{k=1}^M b_k x[n-k] \right) \quad (3.19)$$

Example 3.11

Given a LTI system with difference equation of $y[n] = 0.5y[n-1] + x[n] + x[n-1]$, compute the system output $y[n]$ for $0 \leq n \leq 50$ with an input of $x[n] = u[n]$. It is assumed that $y[-1] = 0$.

The MAT/OCT code is:

```
N=50;           %data length is N+1
y(1)=1;         %compute y[0], only x[n] is nonzero
for n=2:N+1
y(n)=0.5*y(n-1)+2; %compute y[1],y[2],...,y[50]
                  %x[n]=x[n-1]=1 for n>=1
end
n=[0:N];        %set time axis
stem(n,y);
```

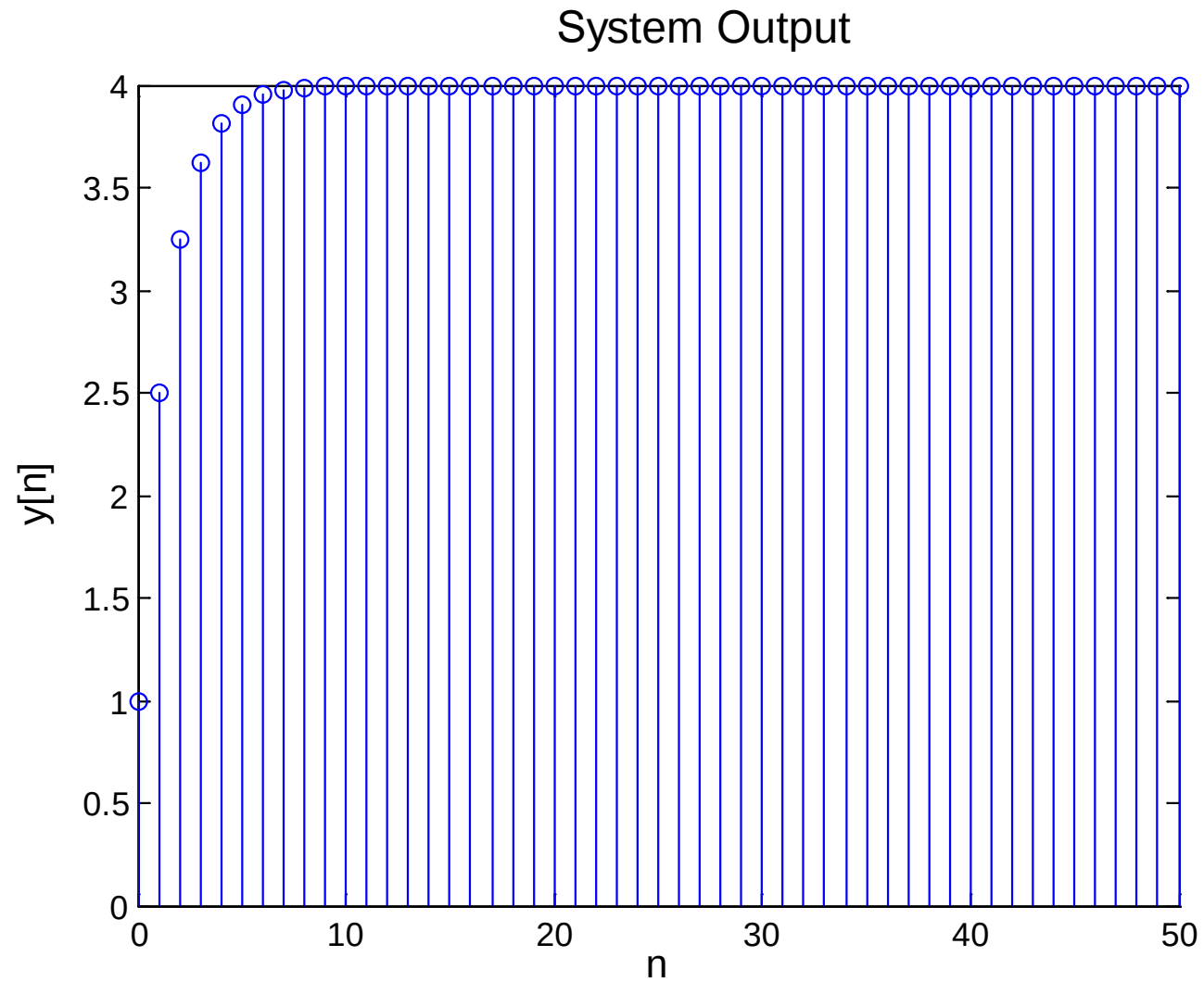


Fig. 3.6 Output generation with difference equation

Alternatively, we can use the MAT/OCT command `filter` by rewriting the equation as:

$$y[n] - 0.5y[n - 1] = x[n] + x[n - 1]$$

The corresponding MAT/OCT code is:

```
x=ones(1,51);           %define input
a=[1, -0.5];           %define vector of a_k
b=[1, 1];              %define vector of b_k
y=filter(b,a,x);        %produce output
stem(0:length(y)-1,y)
```

The x is the input which has a value of 1 for $0 \leq n \leq 50$, while a and b are vectors which contain $\{a_k\}$ and $\{b_k\}$, respectively.

Energy Signals

- A signal can be categorized into energy signal or power signal.
- An **energy signal** has a finite energy, $0 < E < \infty$.
- In other words, energy signals have values only in the limited time duration.
- For example, a signal having one square pulse is energy signal.
- A signal that decays exponentially has finite energy, so, it is also a energy signal.
- The power of an energy signal is 0, because of dividing finite energy by infinite time (or length).

- For a continuous signal:
$$E(t) = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

- For a discrete-time signal:
$$E[n] = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

Power Signals

- A **power signal** has finite power, $0 < P < \infty$.
- A power signal is not limited in time.
- It always exists from beginning to end and it never end.
- For example, sine wave in infinite length is power signal.
 - For a continuous signal: $P(t) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$
 - For a discrete-time signal: $P[n] = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$

Important points:

- A signal cannot be both energy and power signal, simultaneously.
- A signal may be neither energy nor power signal.
- Generally, periodic signals are power signals.
- Generally, aperiodic signals are energy signals.
- Power of energy signal is 0.
- Energy of power signal is ∞ .

Example 3.12

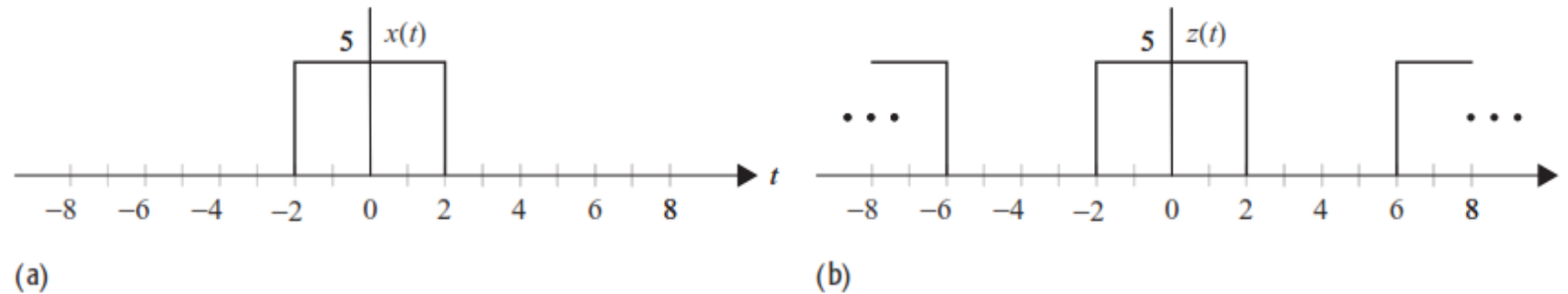


Fig. 3.7 Energy and Power signals

Consider the CT signals shown in Fig. 3.7 (a) and (b). Calculate the instantaneous power, average power, and energy present in the two signals. Classify these signals as power or energy signals.

Solution:

a) The signal $x(t)$ can be expressed as follows:

$$x(t) = \begin{cases} 5 & -2 \leq t \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Instantaneous power } P_x(t) = \begin{cases} 25 & -2 \leq t \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Energy } E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_{-2}^{+2} 25 dt = 100$$

$$\text{Average power } P_x = \lim_{T \rightarrow \infty} \frac{1}{T} E_x = 0;$$

This signal is an energy signal

b) The signal $z(t)$ is a periodic signal with fundamental period 8 and over one period is expressed as follows:

$$z(t) = \begin{cases} 5 & -2 \leq t \leq 2 \\ 0 & 2 \leq |t| \leq 4 \end{cases}$$

$$\text{Instantaneous power } P_z(t) = \begin{cases} 25 & -2 \leq t \leq 2 \\ 0 & 2 \leq |t| \leq 4 \end{cases}$$

$$\text{Average power } P_z = \frac{1}{8} \int_{-4}^{+4} |z(t)|^2 dt = \frac{1}{8} \int_{-2}^{+2} 25 dt = \frac{100}{8} = 12.5$$

$$\text{Energy } E_z = \int_{-\infty}^{+\infty} |z(t)|^2 dt = \infty$$

This signal is a power signal